

Astro Night Planner 1.4.1

Practical manual for astrophotography planning with weather, Moon, object selection, equipment, horizon, framing and local surveys

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Overview

Astro Night Planner combines visibility, twilight, Moon, weather, location, horizon, equipment and Aladin framing in one planning interface. Profiles, equipment, locations and settings are stored locally in the browser. Create an external backup after important changes.

Version 1.4.1 extends local survey support. The Windows solution with control window, tray and autostart remains available. A native Linux/macOS package with browser configuration interface is added.

Planning night, date buttons and Moon

Date buttons summarize the selected night. Weather and Moon values refer to the selected planning window, not automatically to the entire night.

Indicator	Meaning
↑	Moonrise, if it occurs in the relevant night window.
▲	Maximum Moon altitude within the selected planning window .
↓	Moonset. If it is outside the planning window, the planning data says so explicitly.

The planning data may also show the Moon culmination of the night. If it is before or after the planning window, the app adds a corresponding note.

Equipment and Aladin framing

Telescope, reducer/corrector/Barlow and camera determine effective focal length, field of view, pixel scale and the camera frame. In the Aladin view, combinations can be tried temporarily. If a combination is not saved as a setup, it still remains active and can be saved as a new setup.

Example: 700 mm focal length with a 0.80x reducer gives 560 mm effective focal length. With the same camera the field of view becomes larger.

Local surveys and Local Survey Server

Local surveys are local copies of HiPS sky maps. They are served through a local HTTP server, not through file paths. The app and Aladin expect URLs such as `http://127.0.0.1:8765/nsns/ohs8/properties`.

Common principle

All variants serve a shared survey root folder. The app configuration is the same on every operating system:

```
Local survey base URL: http://127.0.0.1:8765/  
Relative path:         nsns/ohs8/
```

The root folder may be `D:\AstroSurveys`, `/home/user/AstroSurveys` or `/Users/user/AstroSurveys`.

Windows variant

On Windows, the **ANP Local Survey Server 1.0** package is recommended. It contains an EXE with control window, tray menu, autostart, background mode and browser configuration interface.

Action	Function
Open interface	Opens the local configuration page.
Start server	Starts the survey file server at the configured base URL.
Stop server	Stops only the survey file server. The helper remains active.
Exit	Stops everything completely and releases the EXE for replacement or deletion.
X in control window	Hides the window to the tray. The program keeps running.

Linux/macOS variant

For Linux and macOS, use **ANP Local Survey Server Linux/macOS 1.0**. It is a native server solution with browser interface. Wine and Python are not required.

Included binaries:

System	File
Linux Intel/AMD 64-bit	<code>bin/anp-local-survey-server-linux-amd64</code>

Linux ARM64	<code>bin/anp-local-survey-server-linux-arm64</code>
macOS Intel	<code>bin/anp-local-survey-server-darwin-amd64</code>
macOS Apple Silicon	<code>bin/anp-local-survey-server-darwin-arm64</code>

Start on Linux

```
chmod +x bin/anp-local-survey-server-linux-amd64
./bin/anp-local-survey-server-linux-amd64 --root "$HOME/AstroSurveys" --port 8765 --open
```

Start on macOS

```
chmod +x bin/anp-local-survey-server-darwin-arm64
./bin/anp-local-survey-server-darwin-arm64 --root "$HOME/AstroSurveys" --port 8765 --open
```

On first launch, macOS may require a security approval because the package is not from the App Store. The configuration interface normally opens at `http://127.0.0.1:8776/`.

Autostart

On Linux, use a `systemd --user` service. The package contains `templates/anp-local-survey-server.service`. On macOS, use a LaunchAgent. The package contains `templates/de.deepskyastrophoto.anp-local-survey-server.plist`.

Python fallback

The Python package remains available as a transparent fallback or expert solution. It is useful if a user prefers a small script or if the native variant does not start on a special platform.

Folder structure

Folder below the root folder	Relative path in Planner	Meaning
<code>nsns/ohs8</code>	<code>nsns/ohs8/</code>	NSNS OIII/H-alpha/SII composite

<code>nsns/halpha8</code>	<code>nsns/halpha8/</code>	NSNS H-alpha
<code>nsns/oiii8</code>	<code>nsns/oiii8/</code>	NSNS OIII
<code>nsns/sii8</code>	<code>nsns/sii8/</code>	NSNS SII
<code>nsns/hbr8</code>	<code>nsns/hbr8/</code>	NSNS H-beta/red, if available
<code>nsns/rgb8</code>	<code>nsns/rgb8/</code>	NSNS RGB, if available

HiPS content and tile format

A HiPS folder must contain a `properties` file. For display, `Norder...`, `Dir...` and `Npix...` are also required. The `properties` file contains `hips_tile_format`. If it says `png`, PNG tiles must be used; if it says `jpeg`, JPEG tiles must be used.

Fallback and diagnostics

The app first checks the local source when local use and local preference are enabled. If reachable, the local URL is passed to Aladin. If it is unavailable and online fallback is enabled, the online source is used. If fallback is disabled, an error message is shown.

Important: The **Open Local Survey Server** button cannot start a local program. It only opens the configuration interface if the matching server is already running.

Weather, cloud model and external sources

Weather, cloud model, Meteoblue, Windy, Ventusky and other external reference sources support the decision whether a night is suitable for astrophotography. External services require an internet connection.

Backup, PWA and troubleshooting

The app uses local browser data. Create regular full backups. Local survey requests to `127.0.0.1`, `localhost` and private LAN

addresses are not intercepted by the PWA cache but passed directly to the local server.

- If local surveys do not work, first check `http://127.0.0.1:8765/nsns/ohs8/properties` in the browser.
- Before replacing the Windows server package, use **Exit** in the control window or tray menu.
- If Linux/macOS should start in the background, configure systemd user service or LaunchAgent.

Glossary

Term	Explanation
HiPS	Hierarchical Progressive Survey, a tiled sky-map format.
properties	Control file of a HiPS survey with tile format and resolution.
Norder	Folder structure of HiPS tiles by resolution order.
Fallback	Switching to the online source if the local source is not reachable and fallback is enabled.

Astro Night Planner 1.4.1 - User manual.